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Industrial Internship Report on

"GroceryPro: Online Grocery Store Management System"

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Executive Summary

This report presents the work completed during the Industrial Internship organized by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). During the internship, a real-world project was assigned and executed within a six-week timeline. The assigned project for this report is "GroceryPro: Online Grocery Store Management System", a full stack web application developed using Java Servlets, JSP, HTML, CSS, Apache Tomcat, and MySQL. The application was designed to provide users with a seamless online grocery shopping experience that includes registration, login, product browsing, cart management, delivery details, payment simulation, order placement, profile access, and order tracking. An admin module was also implemented for product management, image-based product catalog handling, and order status updates. The internship provided practical exposure to solving an industrial problem statement through design, implementation, testing, debugging, and refinement. It significantly strengthened understanding of full stack development, deployment flow, database integration, and user-oriented interface design.

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Preface

The six-week industrial internship was structured to bridge the gap between academic learning and practical industrial implementation. The project was not merely a classroom exercise; instead, it required designing and developing a working web-based solution with industrial relevance. The internship introduced a disciplined workflow including requirement understanding, system planning, user interface design, backend logic development, database connectivity, debugging, testing, and report preparation. This made the learning journey highly practical and meaningful.

The selected project, GroceryPro: Online Grocery Store Management System, was planned as a complete online shopping platform for grocery items. The system was designed to support customer registration and login, image-based product browsing, product search and category filtering, cart functionality, delivery details collection, payment simulation, order placement, order history, profile details, and admin controls. During the internship, the project evolved from a basic static interface into a complete full stack solution with premium user interface improvements and an image-driven catalog.

The internship provided a valuable opportunity through upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd (UCT). The program was planned in such a way that each week concentrated on a practical milestone: project setup, authentication, dashboard and cart, checkout and order flow, admin features, and interface enhancement. This phased structure helped in developing the project step by step while understanding the purpose of each layer in a web application.

My overall experience during the internship was highly productive. I learned how to solve implementation issues systematically, how to integrate frontend and backend components, and how to maintain consistency in project structure and coding flow. I would like to thank the mentors, guides, internship coordinators, upskill Campus team, The IoT Academy, and UCT for providing this opportunity and the required support. To my juniors and peers, I would strongly recommend taking such internships seriously because they provide practical exposure, improve confidence, and make academic knowledge more industry-ready.

Introduction

About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology company working in the domain of digital transformation, industrial IoT, software systems, automation, and sustainability-oriented solutions. The company focuses on delivering industrial value by combining technology with practical return on investment. Its work spans multiple advanced technology areas such as Internet of Things (IoT), cyber security, cloud computing, machine learning, communication technologies, Java full stack development, Python-based systems, and modern frontend solutions.

One of the major takeaways from the company profile is that UCT does not approach technology as an isolated academic concept. Instead, it uses technology to create deployable solutions for industrial monitoring, production optimization, factory analytics, and predictive maintenance. This orientation toward real use cases made the internship highly relevant because it emphasized the need to design solutions that are functional, maintainable, and aligned with business requirements rather than being limited to theoretical demonstration.

UCT's technology ecosystem includes IoT platforms for device connectivity, data analytics dashboards, reporting and visualization systems, industrial communication support, and integration mechanisms for enterprise tools. Such a background strongly influenced the internship program because it encouraged interns to think about practical system flow, modular design, and overall usability. Even though my assigned project was a web application for grocery management, the industrial perspective of UCT helped me focus on solution architecture, testing, and continuous refinement.

The exposure to an industrial partner like UCT broadened my understanding of how software projects are developed in structured phases. From requirement identification to UI refinement and backend connectivity, every step of the GroceryPro project was approached with attention to industrial applicability. This made the internship more meaningful than a simple coding assignment and highlighted the importance of clean design, reliable user workflows, and future scalability.

About upskill Campus

upskill Campus (USC), along with The IoT Academy, played a key role in the smooth execution of the internship. It acted as the learning and coordination platform through which the project, weekly progress reporting, mentorship flow, and deliverable structure were organized. The platform aims to support career development by making industrial learning more accessible, measurable, and outcome-oriented.

USC's contribution was not limited to providing a project statement. It also shaped the internship into a weekly progression model so that each phase contributed toward a final working solution. This kind of structured internship support helped me stay consistent, document work week by week, and convert gradual progress into a complete project. It also reinforced the importance of combining hands-on development with regular reporting and reflection.

The IoT Academy, functioning as the EdTech division associated with the program, complements this ecosystem by connecting learners with industry-focused certification, training, and implementation exposure. The collaboration between USC, The IoT Academy, and UCT made the internship highly relevant to career development because it provided a practical project, guided progression, and a final outcome that can be demonstrated as a full stack implementation.

Objective

- To gain practical industrial exposure in full stack web development.
- To understand how real-world problems are converted into software solutions.
- To design and implement a complete web application using frontend, backend, and database technologies.
- To improve problem-solving, debugging, deployment, and user interface design skills.
- To strengthen job readiness through a demonstrable industry-oriented project.

Reference

[1] upskill Campus internship material and weekly guidance documents.

[2] UniConverge Technologies Pvt Ltd internship sample report and problem statement format.

[3] Official documentation of Java Servlets, JSP, Apache Tomcat, MySQL, and JDBC.

Glossary

Terms	Acronym / Meaning
JSP	JavaServer Pages
JDBC	Java Database Connectivity
DBMS	Database Management System
UI	User Interface
UX	User Experience
CRUD	Create, Read, Update, Delete
HTTP	HyperText Transfer Protocol
UCT	UniConverge Technologies Pvt Ltd

Problem Statement

The assigned problem statement was to design and implement a full stack grocery shopping web application that can simulate an online grocery ordering experience for customers while also providing administrative controls for the store manager. The system needed to support a realistic end-to-end flow rather than isolated pages. The final solution was expected to be a working application with frontend pages, backend processing, database interaction, and report documentation.

The project was framed around a practical use case: a user should be able to register, log in, browse products, view real product images, add products to a cart, enter delivery details, choose a payment option, place an order, and review order history. In parallel, an admin should be able to add products, upload images, edit product data, delete products, and update order status. This made the problem statement sufficiently broad to cover important aspects of web application engineering such as authentication, session management, CRUD operations, and workflow design.

The main challenge of the problem statement was that the solution needed to combine usability and implementation complexity. It was not enough to make pages look correct; the application also had to interact reliably with the database, route requests properly through servlets, maintain session state, and support data persistence. Hence, the project required both design thinking and technical integration.

Existing and Proposed Solution

Traditional solutions for small grocery stores are often offline, manual, or semi-digital. In many cases, customers either visit the store physically or communicate their needs through calls and messages. Even when a digital record exists, it is commonly limited to billing or inventory management rather than a full customer ordering system. Such approaches reduce convenience, make order tracking difficult, and provide limited visibility to both the store owner and the customer.

Existing e-commerce grocery solutions available in the market are feature-rich but are often complex and resource-heavy for a small-scale academic implementation. They usually include large infrastructure, extensive payment gateways, advanced inventory engines, recommendation systems, and delivery partner integration. While these solutions are effective at scale, they are not ideal as a learning implementation if the goal is to understand and build the complete stack from the ground up.

The proposed solution in this internship was GroceryPro, a modular grocery web application built using HTML, CSS, JSP, Java Servlets, Apache Tomcat, and MySQL. The system was designed to include the most important business features in a manageable architecture: signup, login, product display, real product images, search and category filtering, cart management, delivery details, payment simulation, order placement, order history, profile page, admin product management, and admin order updates.

A major value addition of the proposed solution is that it is image-based, session-aware, and admin-manageable. Instead of using placeholder icons or static datasets, the system supports product image mapping, premium user interface design, and database-driven rendering. This makes the project more aligned with modern grocery applications and improves both the user experience and the demonstration value of the project.

Code submission (GitHub link): [Placeholder - to be updated by student]
Report submission (GitHub link): [Placeholder - to be updated by student]

Proposed Design/ Model

The GroceryPro application was designed as a layered full stack system where each user action passes through a clear sequence: frontend interface -> servlet controller -> database operation -> response rendering. This design keeps responsibilities separated and makes the application easier to debug and extend.

High Level Diagram / Architecture

Figure 1: HIGH LEVEL ARCHITECTURE OF THE SYSTEM

User Layer	Customer / Admin	Interacts through browser pages such as index, login, signup, dashboard, cart, admin pages
Presentation Layer	HTML, CSS, JSP	Shows forms, cards, order history, cart tables, profile, and premium landing pages
Controller Layer	Java Servlets	Processes signup, login, add-to-cart, remove-from-cart, place-order, logout, edit-product, delete-product, update-status
Data Layer	JDBC + MySQL	Stores user data, product data, cart items, and order information
Deployment Layer	Apache Tomcat	Hosts and runs the Java web application locally

Low Level Diagram / Module Details

The low-level design consists of multiple modules working together. The authentication module contains SignupServlet and LoginServlet for credential processing and session creation. The product module contains admin product creation, product listing on the dashboard, image binding, and edit/delete operations. The cart module contains AddToCartServlet and RemoveFromCartServlet. The checkout module includes delivery details collection, payment selection, simulated order success handling, and order insertion into the database. The order module contains user order history and admin order status update. The profile module summarizes customer information and order activity.

Each module was implemented with a focus on practical maintainability. JSP pages handle rendering and forms, while servlet classes handle business logic and request processing. JDBC connections are used to perform insert, update, delete, and select queries. Session attributes are used to ensure that only logged-in users can access protected pages such as dashboard, cart, and profile.

Interfaces

- User Signup Interface - collects name, email, password, phone, and address.
- User Login Interface - validates credentials and initiates session.
- Dashboard Interface - displays searchable and filterable product cards with real images.
- Cart Interface - shows selected products, total amount, and remove action.

- Delivery Interface - collects address, city, pincode, mobile, and slot.
- Payment Interface - supports payment simulation using COD, UPI, and card options.
- Admin Product Interface - supports add, edit, delete, and image management.
- Admin Order Interface - supports order monitoring and status update.

Performance Test

Although the project is an academic full stack implementation, performance and reliability considerations were treated as part of the design. The main constraints considered were correctness of request flow, database consistency, session validity, error-free CRUD execution, response usability, and stability of the checkout sequence. The following test plan was used to validate the major workflows.

Test Plan / Test Cases

Test Case	Input / Action	Expected Result	Status
User signup	Enter valid registration details	User record created in users table and success response displayed	Passed
User login	Enter valid email and password	Dashboard opens and session is created	Passed
Add product	Admin submits product name, category, price, and image	Product stored in products table with image_name	Passed
Dashboard display	Open dashboard after login	Products displayed with image, category, price, and add-to-cart button	Passed
Add to cart	Click add to cart on product card	Item inserted into cart table	Passed
Remove from cart	Click remove on cart page	Item removed from cart table	Passed
Checkout flow	Enter delivery details and payment method	Order inserted into orders table and cart cleared	Passed
Order status update	Admin changes order status	Updated status shown in admin orders and order history	Passed

Test Procedure

The testing process followed a sequential execution model. First, user registration and login were verified to ensure that authentication worked and session creation was reliable. After that, admin-side product creation was tested to confirm database insertion and image reference handling. Once products were available, dashboard rendering, category filtering, and search suggestions were validated.

The cart and checkout modules were then tested by adding multiple products, verifying totals, removing items, entering delivery details, choosing payment simulation methods, and confirming order insertion. Admin order updates were validated by changing status values and checking whether the user-facing order

history reflected the same values. The final procedure also included UI inspection, servlet mapping validation, and page-to-page navigation checks.

Performance Outcome

The application produced correct outputs for all primary functional scenarios that were included in the project scope. Customer registration, login, session handling, product rendering, cart operations, checkout flow, order storage, and admin status updates worked successfully after debugging and integration corrections. The outcome demonstrates that the application meets its intended purpose as a complete, database-driven, full stack grocery web solution.

From a performance perspective, the application is appropriate for the internship scope. Since the solution runs on a local Apache Tomcat server with a MySQL backend, response times for form submissions and page rendering are satisfactory for demonstration and academic deployment. The more significant outcome, however, is reliability of the workflow. By the end of development, the project had a stable ordering pipeline, improved user interface consistency, real product images, and clear admin/customer functionality.

My Learnings

The internship significantly improved my practical understanding of full stack web development. I learned how frontend and backend layers communicate through forms, servlet mappings, and database queries. Before this project, many concepts such as session handling, deployment on Tomcat, JDBC operations, and servlet-based request processing were theoretical. Through this project, these concepts became concrete and manageable.

I also learned the importance of debugging in real development. Several issues were encountered during the project, including Tomcat startup problems, servlet mapping errors, missing library configuration, SQL column mismatches, HTML/JSP warnings, path handling for product images, and dashboard rendering issues. Solving these issues taught me how to analyze root causes instead of applying random fixes. This was one of the most valuable aspects of the internship.

Another important learning was user interface refinement. The project evolved from a very basic working interface into a premium-looking application with real images, animated product cards, premium landing pages, better forms, and consistent navigation. This showed me that a successful project is not only technically correct but also visually usable. As a result, I now understand how to balance logic implementation with user experience considerations.

In terms of career growth, this internship has strengthened my confidence in Java web development, database integration, and practical project execution. It has also given me a complete project that can be shown in demos, viva sessions, internship presentations, and portfolio discussions.

Future Work Scope

The current GroceryPro implementation provides a strong functional base, but several advanced enhancements can be taken up in future work. One major extension would be role-based authentication so that customer and admin access can be differentiated more securely. Another future enhancement is proper product image replacement during edit operations, including image upload updates directly from the admin edit page.

The payment system currently simulates payment selection. In future, this can be upgraded with actual payment gateway integration using a sandbox API. Inventory management can also be added so that product quantities reduce automatically after order placement. Another useful scope is category-wise promotions, discount management, order cancellation, invoice generation, and email/SMS notifications.

The platform can further be extended with responsive mobile-first optimization, recommendation logic, product reviews, wish list support, search ranking, and cloud deployment. From an engineering standpoint, the codebase can be improved using MVC structuring, reusable database helper classes, centralized validation, and role-based navigation. These improvements would make the project closer to a production-grade e-commerce application.

Project Links

Code submission (GitHub link): https://github.com/saileshnerusu/upskillcampus

Report submission (GitHub link):
https://github.com/saileshnerusu/upskillcampus/blob/main/GroceryPro_Sailesh_USC_UCT.pdf